# Forensics for Cybercrimes Committed Using Artificial Intelligence

A Digital Forensic Framework for AI-Assisted Cybercrime

**Peter M. Ngugi**
PhD in Computer Science
Zetech University

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# Research Context

Artificial Intelligence is changing the cybercrime landscape.

### Traditional Cybercrime

- Human-controlled systems
- Identifiable devices
- Conventional malware
- Established forensic procedures

### AI-Assisted Cybercrime

- Automated attacks
- Synthetic identities
- AI-generated content
- Autonomous decision-making

# The Forensic Challenge

**AI** → **Cybercrime** → **Evidence** → **Forensics**

*What happened?*

$\Downarrow$

*Who or what caused it?*

$\Downarrow$

*Can we prove it?*

# Formalising Digital Evidence

Let

$$E = \{e_1, e_2, \ldots, e_n\}$$

represent the set of collected digital evidence.

Let

$$A \subseteq E$$

represent evidence that satisfies the authentication criteria.

Therefore:

$$\boxed{A \subseteq E}$$

and

# Forensic Evidence Pipeline

Collection → Preservation → Authentication → Analysis → Attribution

## Core Principle

Evidence must remain traceable from collection through analysis and ultimately to attribution and presentation.

# Research Objectives

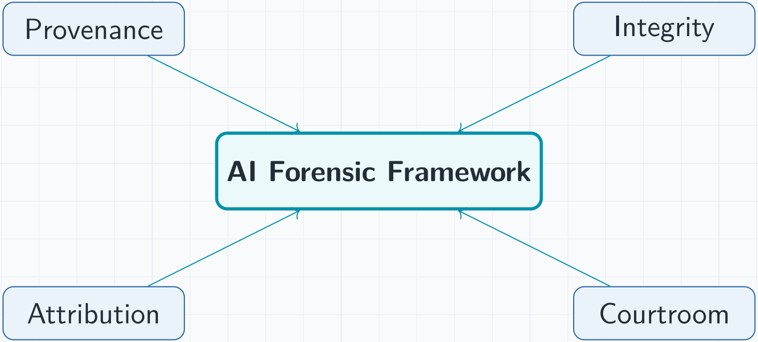
## General Objective

To develop a digital forensic framework for investigating cybercrimes committed using Artificial Intelligence.

**Specific Objectives**

1. Examine AI-assisted cybercrime techniques.
2. Evaluate existing digital forensic approaches.
3. Investigate evidence provenance and integrity.
4. Develop a forensic framework.
5. Evaluate the proposed framework.

# Expected Contribution



# Conclusion

## Central Argument

AI changes not only how cybercrimes can be committed, but also how digital evidence must be interpreted, authenticated and attributed.

$$\boxed{\text{AI} + \text{Cybercrime} + \text{Digital Forensics} + \text{Evidence}}$$

# Thank You

Questions & Discussion

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